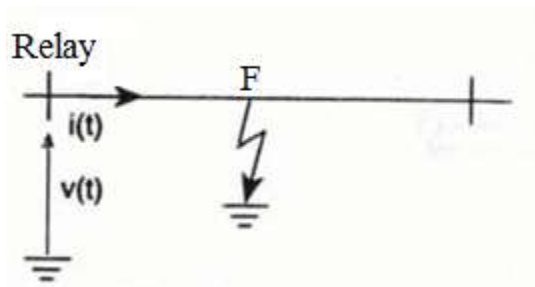
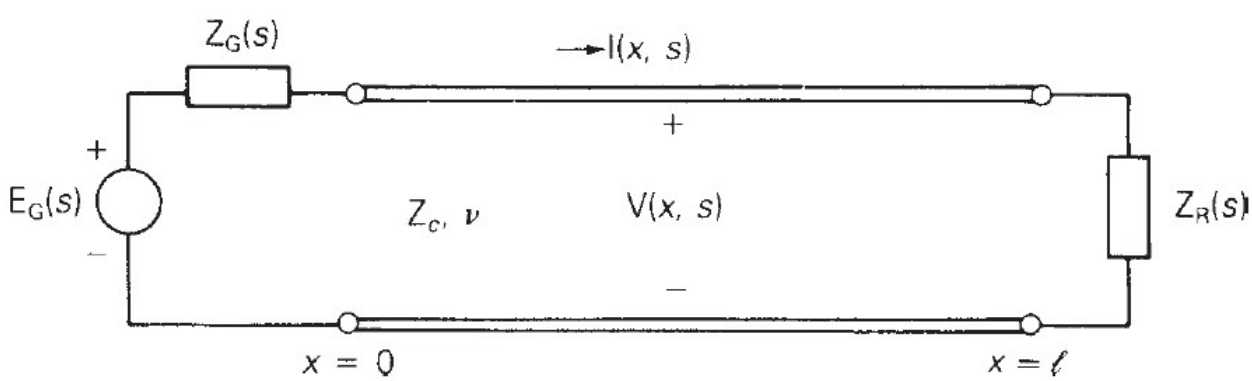
(1) In the following single phase network, assume a fault occurs with Rf=0 at point F. Assume a R-L model for the line. The measured i(t) and v(t) at the relay location in addition to fundamental harmonic include third harmonics as well. Determine how R and L between fault location and the relay location can be estimated using differential equation-based technique.



(2) Referring to the following figure, the receiving end is terminated by ZR=4Zc. The source voltage at the sending end is a step, $e_G(t) = EU_{-1}(t)$ with ZG= Zc/3.. Determine and plot the voltage at the center of the line $v(l/2, t)$ versus time t (represent it based on τ).



(3) Draw the Bewley lattice diagram for Problem 2 and plot $v(l/3, t)$ versus time t for $0 \leqslant t \leqslant 5\tau$. Also plot $v(x, 3\tau)$ versus x for $0 \leqslant x \leqslant l$.